Isha Nilesh Gohel

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Education

University of Massachusetts, Amherst September 2024 - May 2026
MS in Computer Science (GPA: 3.77 / 4.00) *Amherst, MA*

- **Relevant Coursework:** Distributed and Operating Systems, Algorithms in Data Science, Theory and Practice of Software Engineering, Systems in Data Science, Applied Information Retrieval, Reinforcement learning, Advance Machine Learning.

Birla Institute Of Technology and Science, Pilani August 2018 – July 2023
MS in Mathematics and BE in Computer Science (GPA: 8.32 / 10.00) *Hyderabad, India*

- **Relevant Coursework:** Data Structures and Algorithms, Object-Oriented Programming, Database Systems, Computer Networks, Operating Systems.

Technical Skills

Languages: Java, Python, C, C++, JavaScript, React, HTML/CSS, SQL, Bash.
Frameworks and Libraries: PyTorch, TensorFlow, OpenCV, NumPy, Pandas, Dropwizard.
Cloud and DevOps: AWS, GCP, Docker, Kubernetes, Git, Terraform, CI/CD.
Concepts: RESTful APIs, Distributed Systems, Microservices, Caching, Object-Oriented Design, Scalable Systems, Observability/Monitoring (New Relic), Message queues, Testing, Agile/Scrum, Generative AI, RAG, Agentic Systems

Experience

Flipkart Bengaluru, India
Software Development Engineer - I *July 2023 – July 2024*

- Developed **RESTful and gRPC APIs** for post-order tracking features on Flipkart's health commerce platform using Java and MySQL, enabling real-time order visibility for customers and sustaining 50,000+ daily requests with high availability.
- Led the design and automation of a **customer retention pipeline utilizing GCP tools** (Cloud Functions, BigQuery), which directly contributed to a 22% increase in customer retention.
- Provisioned pre-production GCP environments across 6 microservices to execute **NFR testing (load, performance, and reliability)**, ensuring API stability ahead of high-traffic sale events (Flipkart's Big Billion Days).

Software Development Intern *July 2022 – June 2023*

- Built a system to prevent Cash on Delivery (COD) abuse. Implemented per-user order limits using order APIs and risk rules, reducing fraudulent discount exploitation at scale.
- Migrated backend services from AWS to GCP by authoring **Dockerfiles, Cloud Build configs, and Kubernetes manifests**; deployed Java microservices on GCP using Kubernetes and integrated New Relic for performance monitoring and bottleneck identification.

Microsoft Remote
Software Engineering Mentee *June 2021 – July 2021*

- Built VCONNECT, a **full-stack video calling app** with real-time video, audio, screen sharing, and in-call messaging (Microsoft Engage Program).

UST Global Remote
Summer Intern *May 2020 – June 2020*

- Explored approaches to enhance quantum security in Electronic Health Records (EHR) systems.
- Analyzed lattice-based technology as a potential solution for mitigating quantum threats in blockchain networks.

Projects

UMass Marketplace | *FastAPI, React, MongoDB, Firebase, Gemini API*

- To give UMass students a safer alternative to Facebook groups, built a **campus-exclusive full-stack marketplace** featuring secure authentication, AI-assisted listing creation, real-time chat, and community events.
- Developed real-time peer-to-peer messaging to streamline transactions, ensuring a seamless user experience for the student community.

Distributed Image Processing System | *Python, PySpark, Hadoop HDFS, Docker*

- To classify large-scale image datasets faster, built a **distributed inference pipeline** across a multi-node environment, enabling parallel image processing at scale.
- Trained a custom CNN from scratch and fine-tuned ResNet50 for baseline validation, benchmarking both models across **single-node and multi-node configurations**.

Self-Correcting Agentic RAG | *Python, LangGraph, ChromaDB, FastAPI, Groq*

- Architected a **multi-agent RAG** system with LangGraph orchestrating 3 LLM agents (Retriever, Answerer, Critic); self-correcting retrieval via automated query rewriting and adversarial fact-checking between agents.

Publications

- Kumari, D., Yannam, P.K.R., **Gohel, I.N.**, et al.**"Computational model for breast cancer diagnosis using HFSE framework"** *Biomedical Signal Processing and Control*, 86:105121, 2023
  Underlying project : Built a hybrid feature-extraction technique from mammogram images (GLCM, Gabor, LBP) and trained multiple ML classifiers achieving 94% classification accuracy.